Muhammad Zawad Mahmud

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+1 (347) 530-4675 {mm451,mmahmud9}@fordham.edu

Different Links

LinkedIn | **Google Scholar** | **GitHub** | **Website**

Education

M.S. in Computer Science (2025-ongoing)
Fordham University, USA.
CGPA: **4.00/4.00** [**9 credits**]

B. S. in Computer Science and Engineering (2019-23)
North South University (NSU), Bangladesh.
CGPA: **3.71/4.00** [**Mark Percentage: 90%+, Ranked within the top 5%, Graduated with Academic Distinction**]

Work Experience

Graduate Assistant **|** Dept. of Computer & Information Science **|** Aug 2025 – till now

- Actively contribute to research projects assigned by department.
- Maintain 9hours/week office hours to conduct research on assigned project.

Lab Instructor **|** Department of ECE **|** North South University **|** Feb 2025 – Aug 2025

- Maintained 20 hours/week office hours to guide students with problems related to lab courses.
- Teaching and evaluating the lab courses.

Adjunct Lecturer | Department of CSE | Green University of Bangladesh | Feb 2024 – Jun 2025

- Teaching and evaluating courses such as Structure Programming, Data Structures, Database Management System, and Data Communication.

Research Assistant **|** Center for Applied and Computational Science **|** Jan 2023 – Aug 2025

- Participate on various research projects assigned by supervisor.
- Assist undergraduate students on their research activities.

Undergraduate Teaching Assistant **|** North South University **|** Jan 2022 – Nov 2024

- Maintained 20 hours/week office hours to assist students with their course-related problems.
- Evaluating the quizzes and assignments.

Research Interests

Computer Vision **|** Robot Vision **|** Large Language Models**|** AI for Security

Research Records

Journal Articles:

Published:

- A. A. J. Karim, **M. Z. Mahmud**, and R. Khan, "Advanced vision transformers and open-set learning for robust mosquito classification: A novel approach to entomological studies," *PLoS Computational Biology*, vol. 20, no. 12, pp. e1012654–e1012654, Dec. 2024, doi: https://doi.org/10.1371/journal.pcbi.1012654.
- **M. Z. Mahmud**, M. M. Islam, M. M. Molla, M. F. Hasan, and S. Siddiqa, "Multiple-relaxation-time lattice Boltzmann simulation of viscoplastic Bingham nanofluids in a suddenly expanded channel: a systematic numerical study," *Pramana*, vol. 98, no. 3, Aug. 2024, doi: https://doi.org/10.1007/s12043-024-02795-2.

Under Review:

- A Novel Clinical Dataset with XAI-Integrated Ensemble and Large Language Models for Interpretable Dengue Detection: A Comparative Analysis of Reduced-Feature AI Classifiers, Author: **Muhammad Zawad Mahmud**, Md. Shihab Reza, Samiha Islam, and Riasat Khan, Status: Under Review at *Array (Q1; IF: 4.5)*

Conferences:

Published:

- **M. Z. Mahmud**, M. S. Reza, S. R. Alve, S. Islam, and N. Fahmid, "Advance Transfer Learning Approach for Identification of Multiclass Skin Disease with LIME Explainable AI Technique," *2024 27th International Conference on Computer and Information Technology (ICCIT)*, pp. 109–114, Dec. 2024, doi: https://doi.org/10.1109/iccit64611.2024.11021908.
- **M. Z. Mahmud**, S. Islam, S. R. Alve, and A. J. Pial, "Optimized IoT Intrusion Detection using Machine Learning Technique," *2024 IEEE 3rd International Conference on Robotics, Automation, Artificial Intelligence and Internet-of-Things (RAAICON), 2024*, pp. 167–172, doi: https://doi.org/10.1109/raaicon64172.2024.10928532.
- **M. Z. Mahmud**, I. Jahan, S. R. Alve, M. S. Reza, and S. Islam, "Predicting Adolescents' Attitudes Toward COVID-19 Vaccination Using Machine Learning Techniques," *2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, pp. 1–6, Jul. 2025, doi: https://doi.org/10.1109/qpain66474.2025.11172025.
- A. A. J. Karim, **M. Z. Mahmud**, S. Islam, and A. Azam, "Enhancing Multi-Class Disease Classification: Neoplasms, Cardiovascular, Nervous System, and Digestive Disorders Using Advanced LLMs," *2024 27th International Conference on Computer and Information Technology (ICCIT)*, pp. 73–78, Dec. 2024, doi: https://doi.org/10.1109/iccit64611.2024.11022500.
- S. Islam, **M. Z. Mahmud**, S. R. Alve, M. M. U. Chowdhury, and F. I. Oishe, "Deep Learning Approach for Enhancing Oral Squamous Cell Carcinoma with LIME Explainable AI Technique," *2024 27th International Conference on Computer and Information Technology (ICCIT)*, pp. 381–386, Dec. 2024, doi: https://doi.org/10.1109/iccit64611.2024.11022521.

- S. Islam, S. R. Alve, **M. Z. Mahmud**, F. I. Oishe, and M. M. Khan, "Artificial Intelligence Driven Intrusion Detection for SDN Environments," *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, pp. 1–6, Feb. 2025, doi: https://doi.org/10.1109/ecce64574.2025.11013409.
- S. R. Alve, **M. Z. Mahmud**, S. Islam, Md. A. Chowdhury, and J. Islam, "Resource-Efficient Machine Learning Approaches for Multi-Class Threat Detection in IoT Environments," *2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, pp. 1–6, Jul. 2025, doi: https://doi.org/10.1109/qpain66474.2025.11171769.
- F. Elias, M. S. Reza, **M. Z. Mahmud**, S. Islam, and S. R. Alve, "Interpretable Machine Learning for Osteoporosis Risk: A Comparative Evaluation of Predictive Models and XAI Analysis," *2025 International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, pp. 1–6, Jul. 2025, doi: https://doi.org/10.1109/qpain66474.2025.11172213.
- A. J. Pial, M. A. U. I. Sajid, A. Mubin, **M. Z. Mahmud**, M. F. Mridha, and A. Jubair, "Optimized Malaria Identification through Transfer Learning Appoach," *Proceedings of the 3rd International Conference on Computing Advancements*, pp. 719–724, Oct. 2024, doi: https://doi.org/10.1145/3723178.3723273.

Submitted:

- Systematic Evaluation of Novel View Synthesis for Video Place Recognition, Author: **Muhammad Zawad Mahmud**, Samiha Islam, and Damian Lyons, Status: Submitted at *International Conference on Intelligent Robots and Systems (IROS)*. Preprint available at https://doi.org/10.48550/arXiv.2603.05876

Ongoing:

- Privacy Leakage in Federated Learning: A Membership Inference Attack Analysis with Differential Privacy Defenses, Author: Muhammad Zawad Mahmud and Mohamed Rahouti.

Academic Projects

- Food Delivery App using C programming language
- Course Registration and Billing System using Java programming language
- Graph Adjacency List Implementation using Java
- Air Traffic Control Simulation using Queue and Priority Queue
- Data Searching and Sorting from CSV file using C++, Python and Shell Script
- 12 Bit CPU Design in Logisim
- IoT Device's Intrusion Detection using Machine Learning Methods
- Malaria Prediction using Deep Learning Techniques
- Multi Class Skin Disease Prediction using Deep Learning
- Advance MapReduce Analytic Pipeline
- Benchmarking Kafka with Zookeeper and Kraft

Technical Skills

- **Programming Languages:** C | C++ | Python
- **Frameworks:** PyTorch | Keras | TensorFlow
- **Operating Systems (OS):** Linux (Ubuntu) | Windows
- **Software Tools:** VS Code | Google Colab | Jupyter Notebook
- **Soft Skills:** Teamwork | Communication | Networking

Academic Awards

- Merit Based Scholarship in graduate studies.
- Financial Aid (75%) based on academic performance in undergraduate studies.
- Magna Cum Laude based on undergraduate CGPA (e.g., CGPA >=3.65)

Standardized Test Scores

- TOEFL – **107** (L-29, R-28, S-24, W-26)
- GRE – **321** (Verbal-159, Quantitative-162), AWA-4.5

Research Training and Achievements

- Training On: Applied Machine Learning & Deep Learning
  Supervisor: Dr. Mahdy Rahman Chowdhury
- Reviewer of Internet of Things & Cyber-Physical Systems, Next Material Journal, Meteorological Applications, and Franklin Open.
- Reviewer of International Conference on Intelligent Data Analysis and Applications (IDAA 2025).
- Reviewer of 2026 IEEE 27th International Conference of Young Professionals in Electron Devices and Materials (EDM).